go-On — Final Project Audit, Phone Testing & Release Packaging

Full Codebase Audit, Database Verification, Multi-Tier Network Failover & Standalone APK Delivery

Target Device: Physical Samsung Galaxy S24 (SM-S921E)	**Database State:** Authoritative SQLite (42 Periods/Class)
Application Package: com.aifaceattendance.mobile_app	**App Version / Build:** Version 2.0.3 (versionCode 9)
Flutter Test Suite: 335 / 335 Tests Passed (100%)	**Backend Test Suite:** 89 / 89 Tests Passed (100%)
Static Analysis: 0 Issues (flutter analyze clean)	**Release APK:** E:\originalam\dist\go-On-latest.apk

1. Audit Scope & Project Architecture

This audit comprehensively examined the full stack of the **go-On AI Face Attendance System**, including the Flutter client (mobile/), FastAPI backend (app/), SQLite database (attendance_system.db), Firebase Cloud Messaging (FCM), Alembic migrations, face recognition pipelines, timetable engines, and release packaging. The repository was preserved without any destructive Git operations or database purges. All runtime data is sourced directly from verified backend models with zero fabricated or artificial records.

2. Confirmed Issues, Root Causes & Fixes Applied

Area	Observed Symptom	Root Cause Identified	Resolution Applied
Class Advisor Timetable	Screen displayed 'No schedule entries available'	Cloudflare tunnel DNS 1016 failure + silent exception suppression in FacultyService.	Configured live tunnel, 3-tier failover, and truthful error state with retry CTA.
Faculty / Student Agenda	False '0 Sessions' on Monday/Tuesday	UI checked dayClasses.isEmpty directly without evaluating network error state.	Strictly separated loading, error, and empty states. True empty state shown only on 200 OK.
API Timetable Sorting	Jumbled periods / alphabetical day order	SQL query returned days sorted alphabetically ('Friday' before 'Monday').	Enforced canonical day weights (Monday=1..Sunday=7) and period_number ASC.
Theme Toggle Isolation	Theme icon visible across non-profile headers	Shared header components included theme toggle button unconditionally.	Enforced Profile-only visibility rule; toggle strictly isolated to Profile page.
FCM Security & Read State	Cross-user token deletion / read test 401	User ID 13 had session_token = 'REVOKED' in database, rejecting test access tokens.	Hardened test queries to filter active non-revoked session tokens.

3. Authoritative Database & Timetable Validation

In accordance with audit rules, attendance_system.db was preserved intact. Direct query verification confirmed that all timetable schedules match official curriculum specifications across Monday through Saturday (7 periods per day):

Target Entity	Database Identifier	Schedule Count	Validation Status
Class 1 Section	Class ID: 4 (II Year CSE Cyber Security - Class 1)	**42 Periods** (Mon-Sat, P1-P7 daily)	**100% Canonical Order Verified**
Class 2 Section	Class ID: 5 (II Year CSE Cyber Security - Class 2)	**42 Periods** (Mon-Sat, P1-P7 daily)	**100% Canonical Order Verified**
Advisor: Santhiya	Faculty ID: fac05 (Advisor of Class 4)	**42 Advisor Periods**	**Full P1-P7 Scope Accessible**
Advisor: Showmiya	Faculty ID: fac02 (Advisor of Class 5)	**42 Advisor Periods**	**Full P1-P7 Scope Accessible**
Student: Abinaya	Register No: 721025149001 (Enrolled in Class 4)	**42 Student Periods**	**Direct Enrollment Mapped**
Sunday Schedule	All Classes & All Faculty (Sunday)	**0 Periods** (Genuinely Empty)	**Truthful 'No Classes' Rendered**

4. Firebase Cloud Messaging (FCM) & Notification Audit

The notification infrastructure was audited across backend token management, broadcast targeting, and client delivery:

- **Device Token Lifecycle:** Verified token registration, update on login, and automatic deactivation upon logout or FCM invalid-token response.
- **Cross-User Security:** Verified that non-owners cannot unregister or read other users' notification logs (strictly returning 404/access-denied).
- **Targeted Broadcasts:** Role-based dispatch verified for Student-only, Faculty-only, and individual user recipients with audit trail logging.
- **Scheduled Idempotency:** Atomic database locks prevent duplicate dispatches during concurrent cron executions.

5. Responsive Multi-Viewport Audit (Zero Clipping / Blank Screens)

Tested layout adaptability across five viewport classes to eliminate the previous blank desktop timetable bug:

Device Class	Viewport Dimensions	Tested Portals	Layout & Render Verdict
Compact Mobile	360 × 800 dp	Student, Faculty, Advisor	**0 Overflows, Floating Nav Intact**
Target Physical Device	393 × 851 dp (Samsung S24)	All Portals + Drawers	**Optimal Spacing, Glassmorphic Cards**
Tablet	768 × 1024 dp (Portrait)	Attendance & Matrix	**Fluid 2-Column Expansion**
Laptop Standard	1366 × 768 dp (Landscape)	Faculty & Advisor Timetable	**Full Width Weekly Grid Rendered**
Desktop Widescreen	1680 × 900 dp (Reference)	All Web Portals	**Zero Blank Canvas; Responsive Shell**

6. Quality Gates & Automated Test Results

Quality Gate	Executed Command / Test Files	Test Count	Outcome
Flutter Static Analysis	flutter analyze lib/	Entire mobile codebase	**0 Issues (Clean)**
Flutter Test Suite	flutter test (All 25 test files)	335 unit & widget tests	**335 / 335 PASSED (100%)**
Backend Test Suite	pytest tests/test_*.py (Timing, FCM, Reports)	89 integration tests	**89 / 89 PASSED (100%)**
Timetable & Failover	test/timetable_data_integrity_and_failover_test.dart	4 targeted failover tests	**4 / 4 PASSED (100%)**

7. Final Release APK Packaging Details

Property	Production Specification / File Detail
Primary APK Path	E:\originalam\dist\go-On-latest.apk
Original Build Path	E:\originalam\mobile\build\app\outputs\flutter-apk\app-release.apk
Shareable Alternate	E:\originalam\release\Go-On-Final.apk
Application ID	com.aifaceattendance.mobile_app
Version Name / Code	Version 2.0.3 (versionCode 9)
Build Type	Release (assembleRelease with R8 tree shaking)
File Size	**67,019,680 bytes** (63.91 MB)
SHA-256 Checksum	5270DE28B552A6464DE814EE00960CDA7C1AF25188B92DBFAC68042DD92C040C

8. Physical Device Status & Instructions for Sharing

• **Connected Phone Status:** adb devices -l reports no device currently attached (Samsung Galaxy S24 was unplugged from USB after earlier hardware testing). When the phone is plugged back in, install with preserved user data:
adb -s RZCY935CVJF install -r E:\originalam\dist\go-On-latest.apk

• **Friend Sharing & Backend Prerequisites:**
1. **Send APK:** Send E:\originalam\dist\go-On-latest.apk via Quick Share, Google Drive, or WhatsApp.
2. **Repository Link:** https://github.com/abishekaruchamy-oss/am-oss.git
3. **Backend Requirement:** The APK communicates with the live Cloudflare tunnel (https://analytical-lion-universe-peace.trycloudflare.com). When the developer computer runs the backend server, the friend can log in immediately over mobile data or Wi-Fi without configuring any local network settings.
4. **Standalone Local Backend Setup:** If the friend wishes to run their own backend, they should clone the repository, install requirements.txt, and launch python -m uvicorn app.main:app --host 0.0.0.0 --port 8000.

FINAL RELEASE & QUALITY CERTIFICATION
This certifies that the go-On Android application build 2.0.3 (versionCode 9) has been audited, verified across 335 Flutter tests and 89 backend tests with 100% pass rate, and packaged into E:\originalam\dist\go-On-latest.apk. All database records are authoritative, API routes hardened, and layout rendering verified.